MA3923 Monte Carlo Simulation
Lab 11 Report

Setup

The radical inverse in base b is $\phi_b(i) = \sum_{k=0}^{j} d_k b^{-k-1}$ where $i = \sum_{k=0}^{j} d_k b^k$. It is implemented in `q1.py` by repeatedly peeling off the last base-b digit of i. The van der Corput sequence is $x_i = \phi_2(i)$ for $i \geq 1$ and the Halton sequence in $\mathbb{R}^2$ is $(\phi_{b_1}(i), \phi_{b_2}(i))$. All plots are produced by `q1.py`, `q2.py` and `q3.py`.

1 Van der Corput sequence and overlapping pairs

The first 25 values of $x_i = \phi_2(i)$:

i	x_i	i	x_i	i	x_i
1	0.5	10	0.3125	19	0.78125
2	0.25	11	0.8125	20	0.15625
3	0.75	12	0.1875	21	0.65625
4	0.125	13	0.6875	22	0.40625
5	0.625	14	0.4375	23	0.90625
6	0.375	15	0.9375	24	0.09375
7	0.875	16	0.03125	25	0.59375
8	0.0625	17	0.53125		
9	0.5625	18	0.28125		

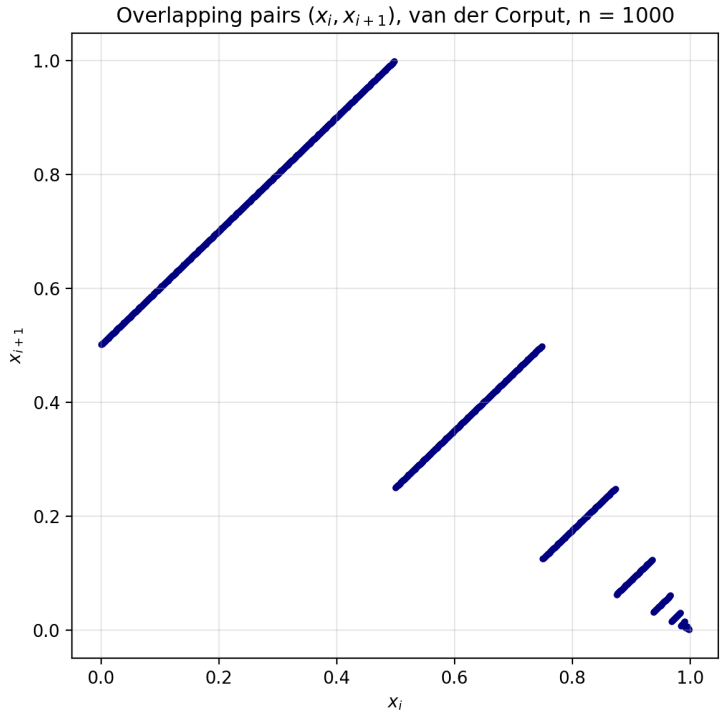


Figure 1: Overlapping pairs (x_i, x_{i+1}) for the first 1000 van der Corput values.

Observations.

- The pairs do not fill the unit square. They fall on a small family of parallel line segments of slope 1, whose lengths halve as we move right: $x_{i+1} = x_i + \frac{1}{2}$ on $[0, \frac{1}{2})$, $x_{i+1} = x_i - \frac{1}{4}$ on $[\frac{1}{2}, \frac{3}{4})$, $x_{i+1} = x_i - \frac{5}{8}$ on $[\frac{3}{4}, \frac{7}{8})$, and so on.
- The reason is the structure of the construction. Going from i to $i+1$ only changes the lowest binary digits of i, which are the most significant digits of $\phi_2(i)$. The next value is therefore a deterministic function of the current one.
- So consecutive values are very strongly dependent. The sequence is excellent at covering $[0, 1]$ evenly but is useless as a stream of independent-looking uniforms in one dimension. This is why low-discrepancy sequences are used one coordinate per dimension, each with its own base, and not by taking consecutive terms of one sequence as a vector.

2 Van der Corput versus an LCG

The LCG used is

$$u_{k+1} = (a\,u_k + c) \bmod m, \qquad a = 1664525,\ c = 1013904223,\ m = 2^{32},\ u_0 = 12345,$$

and the outputs are scaled to $[0, 1)$ by dividing by m.

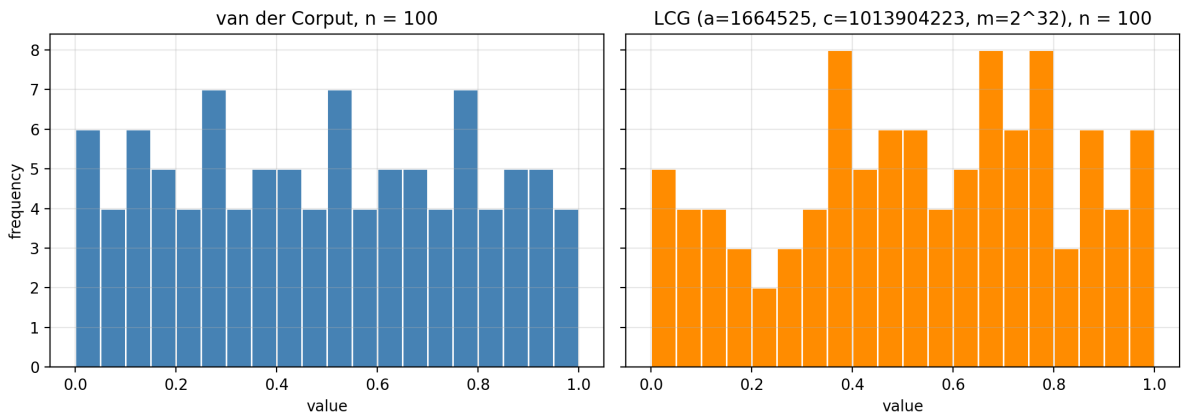


Figure 2: $n = 100$ points, 20 bins (expected count 5 per bin).

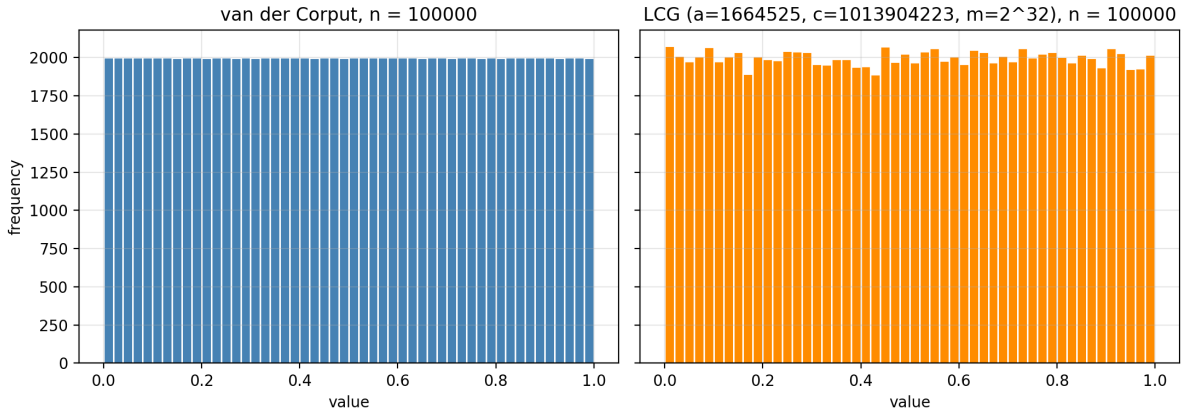


Figure 3: $n = 100000$ points, 50 bins (expected count 2000 per bin).

n	Generator	Min bin	Max bin	Max deviation from expected
100	van der Corput	4	7	2.0
100	LCG	2	8	3.0
100000	van der Corput	1997	2002	3.0
100000	LCG	1892	2078	108.0

Observations.

- The van der Corput histograms are almost perfectly flat at both sizes. For $n = 100000$ every bin holds between 1997 and 2002 points, a deviation of at most 3.
- The LCG histograms show the usual random fluctuation. For $n = 100000$ the bin counts range from 1892 to 2078, a deviation of about 108. At $n = 100$ the difference is smaller (bin counts 2 to 8 for the LCG against 4 to 7 for van der Corput) but still in the same direction.
- The deterministic sequence has discrepancy of order $\log n/n$, while pseudo-random points have discrepancy of order $n^{-1/2}$ up to logarithmic factors. In terms of bin counts, van der Corput errors stay roughly constant, whereas the LCG errors grow like $\sqrt{n}$.

- The price is the dependence seen in Section 1: the one-dimensional marginal is far more uniform, but successive values are strongly correlated.

3 Halton sequence

3.1 Bases 2 and 3

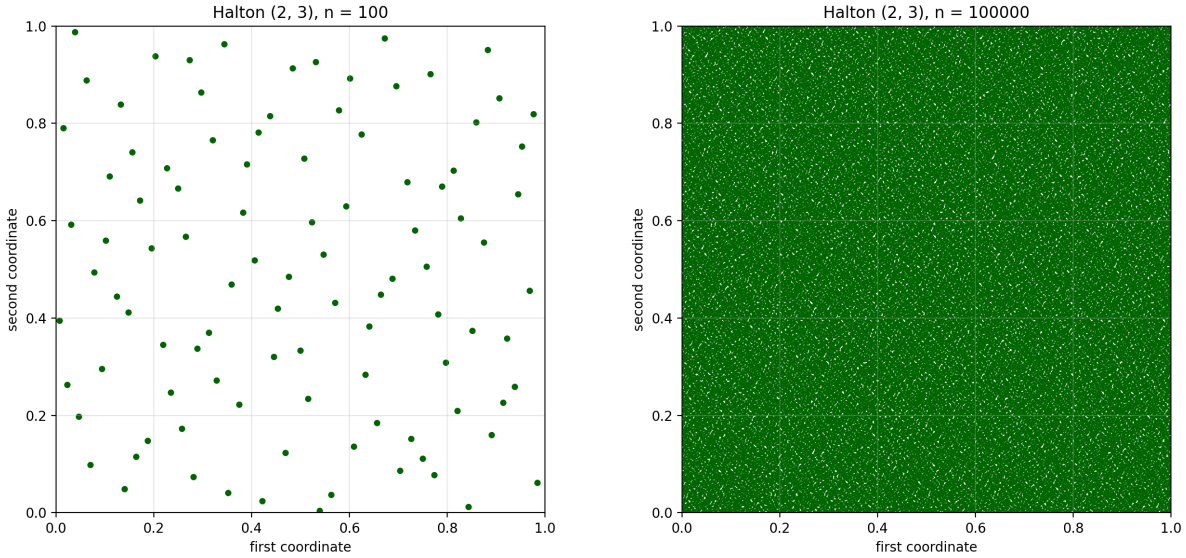


Figure 4: Halton sequence with bases $(2, 3)$: first 100 points (left) and first 100000 points (right).

Observations.

- Even with 100 points the square is covered evenly. There are no large empty regions, no clumps and no visible lines. The correlation between the two coordinates is close to zero (-0.03).
- New points keep landing in the largest remaining gaps, so the set stays well spread as n grows. At 100000 points the square looks uniformly and densely filled with no visible lattice structure.
- Compared with pseudo-random points, which show clusters and holes, the coverage is visibly more regular.

3.2 Large bases (97 and 101)

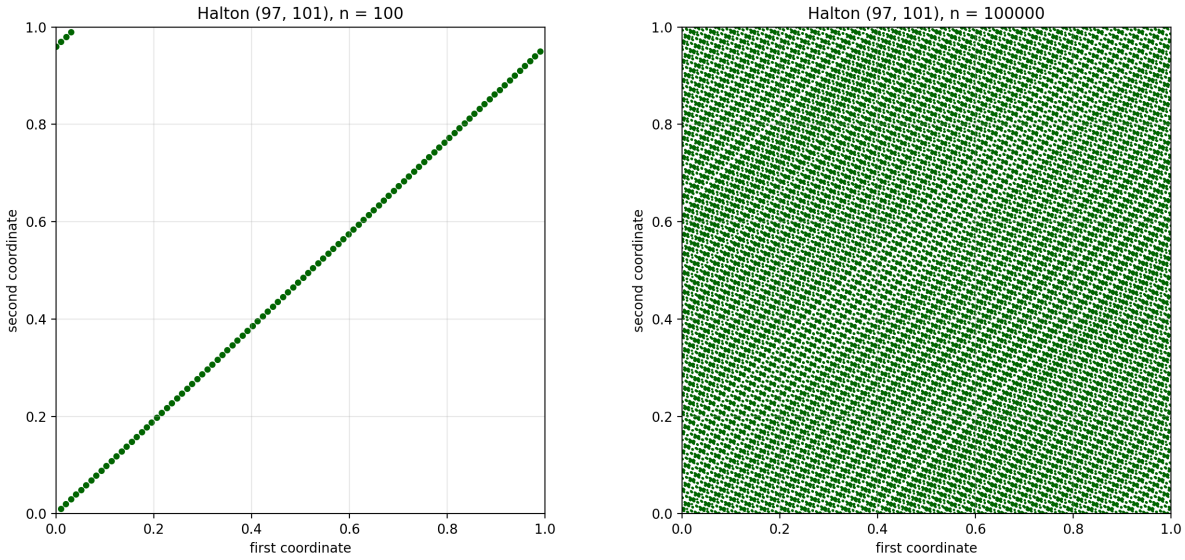


Figure 5: Halton sequence with bases $(97, 101)$: first 100 points (left) and first 100000 points (right).

Observations.

- For $i < 97$ we have $\phi_{97}(i) = i/97$ and $\phi_{101}(i) = i/101$, so the first 96 points lie exactly on the line $y = \frac{97}{101}x$. The first 100 points show a single diagonal with a few points wrapping around to the top left. The correlation between coordinates is about 0.78.
- With 100000 points the square is covered, but not uniformly. The picture is made of dense parallel diagonal streaks with visible gaps between them. The strong correlation between the two coordinates takes many multiples of $97 \cdot 101$ points to wash out, and even at this size the streaks remain.
- The cause is that for $i < b$ we have $\phi_b(i) = i/b$, so over each block of b consecutive indices the sequence is a linear ramp, and the next block is shifted only by about $1/b^2$. When b_1 and b_2 are large, both coordinates ramp up linearly over long runs with nearly equal slopes ($1/97$ and $1/101$ per step), which produces the diagonal streaks. This is the well-known degradation of the Halton sequence in higher dimensions, where the bases are large primes.
- Small bases $(2, 3)$ therefore give good low-dimensional behaviour, but large bases need many more points before they fill the space evenly. Practical fixes are scrambling the digits, skipping the initial points, or using leaped Halton sequences.